

07/09/2026

(*)

$$\overline{A}BC + A\overline{B}C + A\overline{B}\overline{C} + A\overline{B}C$$

$$= \overline{A}BC + A\overline{B}C + A\overline{B}\overline{C} + A\overline{B}C$$

$$= \overline{A} + \overline{B} + \overline{C} + \overline{A} + \overline{B} + \overline{C} + \overline{A} + \overline{B} + \overline{C} + \overline{A} + \overline{B} + \overline{C}$$

$$\overline{A} + \overline{B} + \overline{C}$$

$$= A + B + \overline{C} + A + \overline{B} + C + \overline{A} + B + C + \overline{A} + \overline{B} + \overline{C}$$

$$\overline{A} + \overline{B} + \overline{C}$$

$$\overline{A} + \overline{B} + \overline{C}$$

(*)

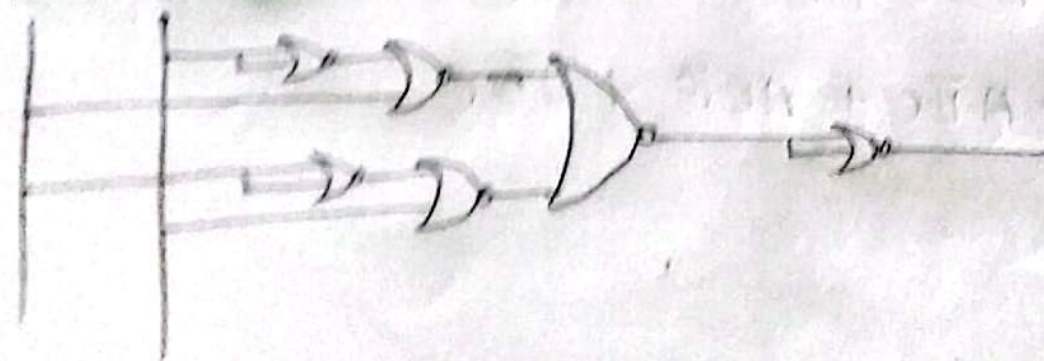
$$\overline{A}B + A\overline{B}$$

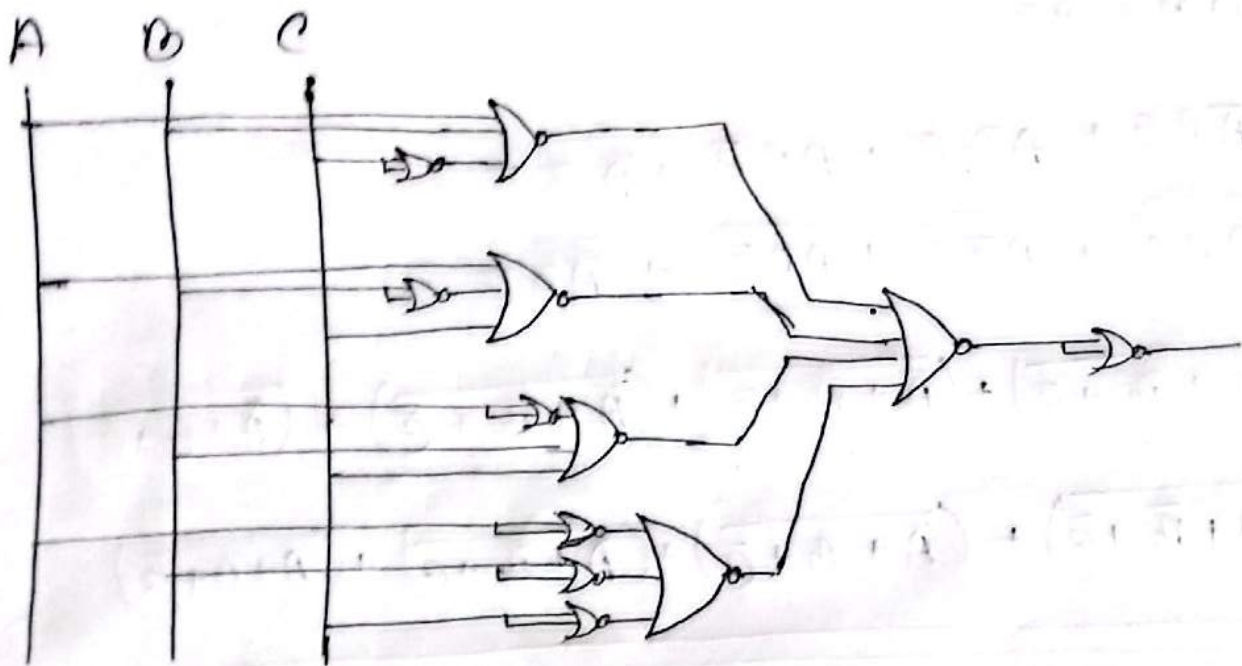
$$= \overline{A}B + A\overline{B}$$

$$= \overline{A} + \overline{B} + \overline{A} + \overline{B}$$

$$= \overline{A} + \overline{B} + \overline{A} + \overline{B} = \overline{A} + \overline{B}$$

A B



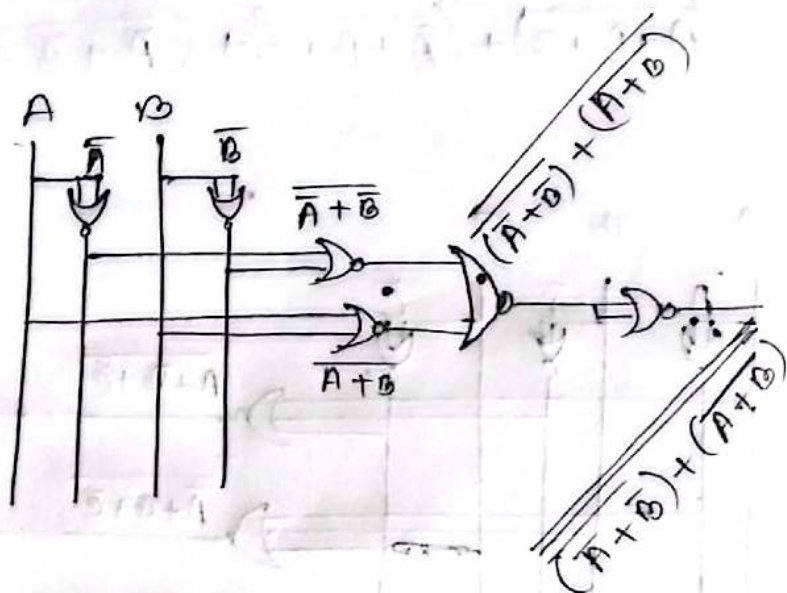


$$(4) AB + \bar{A}\bar{B}$$

$$= \overline{\bar{A}B} + \overline{A\bar{B}}$$

$$= \overline{\bar{A} + \bar{B}} + \overline{A + B}$$

$$= \overline{\bar{A} + \bar{B} + A + B}$$



• Buffer gate: $A \rightarrow \bar{\bar{A}} \rightarrow \bar{\bar{A}} = A$

$$(*) \quad A \oplus B \oplus C$$

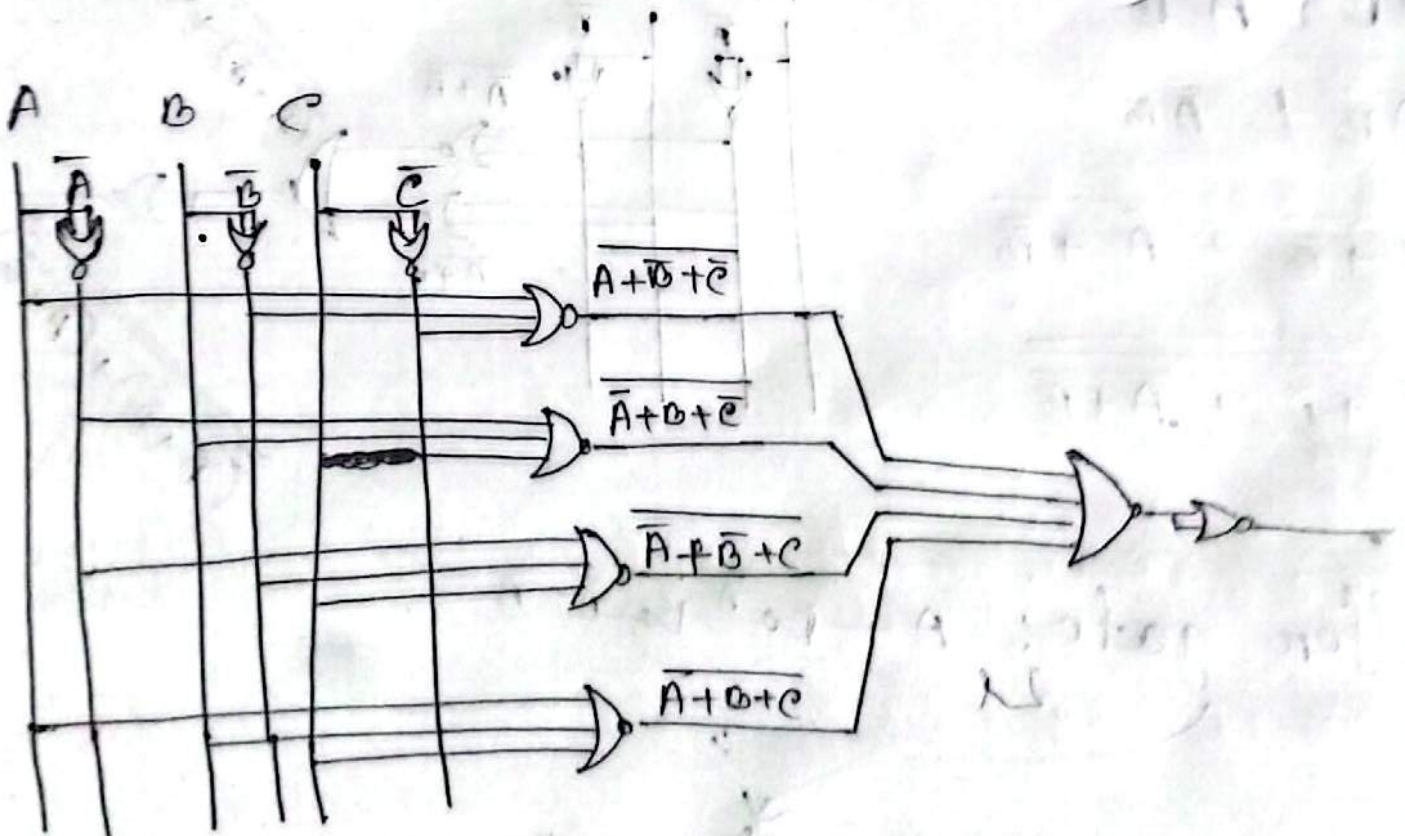
$$= \overline{A}BC + A\overline{B}C + AB\overline{C} + \overline{A}\overline{B}\overline{C}$$

$$= \overline{\overline{A}BC} + \overline{A\overline{B}C} + \overline{AB\overline{C}} + \overline{\overline{A}\overline{B}\overline{C}}$$

$$= (\overline{\overline{A} + B + C}) + (\overline{A + \overline{B} + C}) + (\overline{A + B + \overline{C}}) + (\overline{\overline{A} + \overline{B} + \overline{C}})$$

$$= (\overline{A + B + C}) + (\overline{\overline{A} + B + C}) + (\overline{A + \overline{B} + C}) + (\overline{\overline{A} + \overline{B} + C})$$

$$= \overline{(\overline{A + B + C}) + (\overline{\overline{A} + B + C}) + (\overline{A + \overline{B} + C}) + (\overline{\overline{A} + \overline{B} + C})}$$



HW:

$$(*) Z = \overline{A}C(\overline{A}BD) + \overline{A}B\overline{C}D + A\overline{B}C$$

NOR - Gate - 2 or 3 inputs